

EXERCISE 5: IMAGE PROCESSING PART 2

Multimedia Database SS 26

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Task 1.1: Point Operations

- Assume that we are using an 8-Bits grayscale image. The **HK** point operation is defined as follows:

$$P_{\text{out}} = \alpha P_{\text{in}} + \beta$$

where

P_{in} : Pixel values of the input image

P_{out} : Pixel values of the output image

α known as gain, is a contrast factor

β known as bias, is a brightness factor



Task 1.2: Point Operations

Explain which HK operation could be applied to implement the **image inversion** function.

Solution

$$P_{\text{out}} = - P_{\text{in}} + \text{MaxValue}$$

Where

P_{in} and P_{out} : Sample values of the input and output of a pixel image respectively.



Task 1.3: Point Operations

Which problem could appear, if **HK** is used with a none adapted α and β parameters? Propose a method to deal with these effects.

- Problem:
 - There are no limits on the values of the pixels
- Solution:
 - Clamping; given a sample with a value x such that the sample is defined on n bits (i.e.; $x \in [\min, \max=2^n-1]$ s.t. $\min \geq 0$), a formal definition of the clamping is given as follows:

$$\text{clamp}(x, \min, \max) = \begin{cases} \min & \text{if } x \leq \min, \\ \max & \text{if } x \geq \max, \\ x & \text{otherwise} \end{cases}$$



Task 1.4: Point Operations

4. Let **G** be a grayscale image with minimum pixel value **a** and maximum pixel value **b**. Which HK Operation could be applied to **G** in order to maximize its contrast ratio?
- The maximum contrast is achieved when $HK(a)$ corresponds to the minimum pixel value and $HK(b)$ the maximum pixel value. For a grayscale image:

$$\begin{aligned} HK(a) &= 0, HK(b) = 255 \\ \Leftrightarrow \alpha \times a + \beta &= 0; \alpha \times b + \beta = 255 \end{aligned}$$

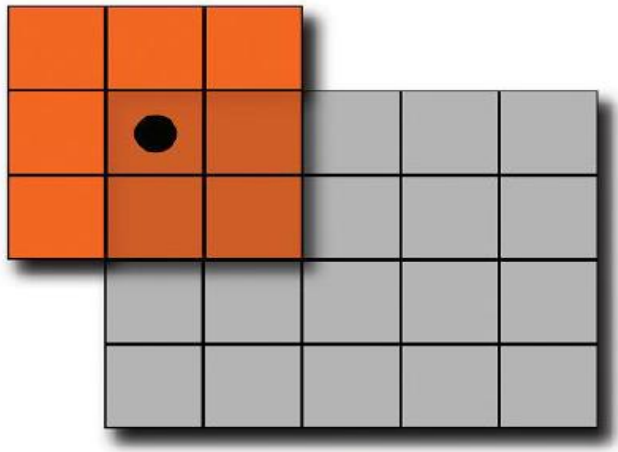
By solving the equation with two unknowns, we will have: $\beta = -a\alpha$,
 $\alpha = 255/(b-a)$



Task 2.1: Linear Filter

- Which problems can occur to edge pixels when using this filter? How can you solve it?

Problem:



The kernel extends beyond the source image boundaries near the image edges.

Solutions:

- Redefine convolution at the edge boundary.
- Padding.

Need help on what is convolution?

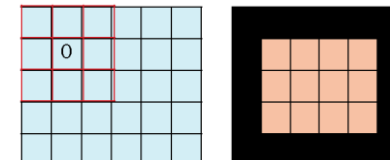
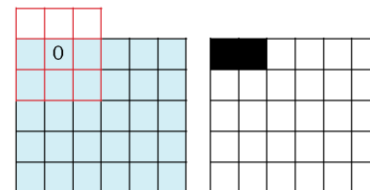
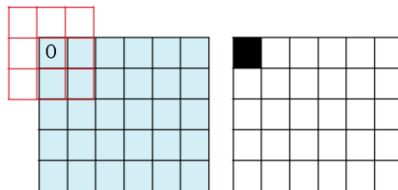
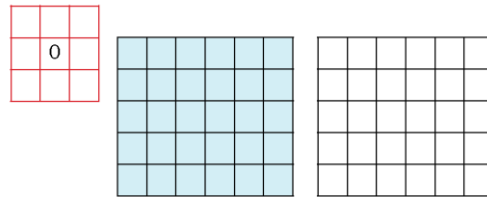
[But what is a convolution?](#)

Source: Youtube 3Blue1Brown

Task 2.1: Linear Filter

Redefine convolution at the edge boundary

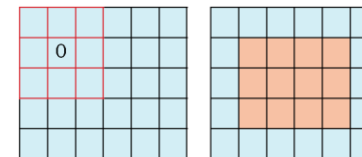
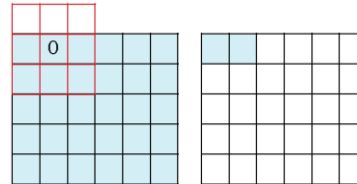
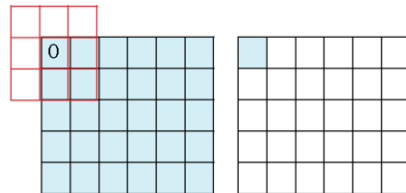
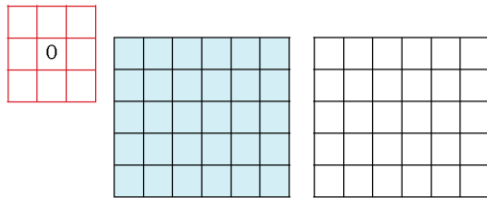
- Convolution is redefined to produce zero when the kernel falls off of the boundary.* If the kernel extends beyond the source image when centered on a sample $I(x, y)$ then the output sample is set to zero.



Task 2.1: Linear Filter

Redefine convolution at the edge boundary

- *Convolution is redefined to produce $I(x, y)$ when the kernel falls off the boundary. If the kernel extends beyond the source image when centered on a boundary sample then the output sample is defined as $I(x, y)$.*



Task 2.1: Linear Filter

- Padding:
 - Zero padding
 - Symmetric Padding

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} \xrightarrow{\text{zero}} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 4 & 0 \\ 0 & 5 & 6 & 7 & 8 & 0 \\ 0 & 9 & 10 & 11 & 12 & 0 \\ 0 & 13 & 14 & 15 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Zero Padding

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} \xrightarrow{\text{symmetric}} \begin{bmatrix} 1 & 1 & 2 & 3 & 4 & 4 \\ 1 & 1 & 2 & 3 & 4 & 4 \\ 5 & 5 & 6 & 7 & 8 & 8 \\ 9 & 9 & 10 & 11 & 12 & 12 \\ 13 & 13 & 14 & 15 & 16 & 16 \\ 13 & 13 & 14 & 15 & 16 & 16 \end{bmatrix}$$

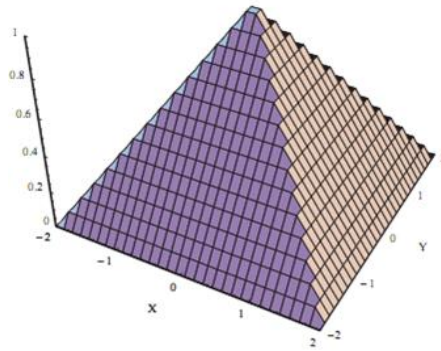
Symmetric Padding

Images from: https://www.uio.no/studier/emner/matnat/ifi/INF2310/v17/undervisningsmateriale/slides_inf2310_s17_week06.pdf

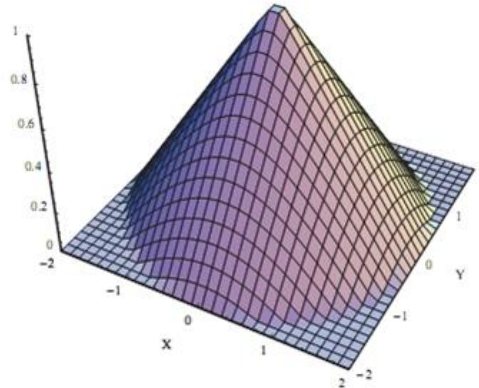
Task 2.2: Moving Average Filter

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} * \frac{1}{25}$$

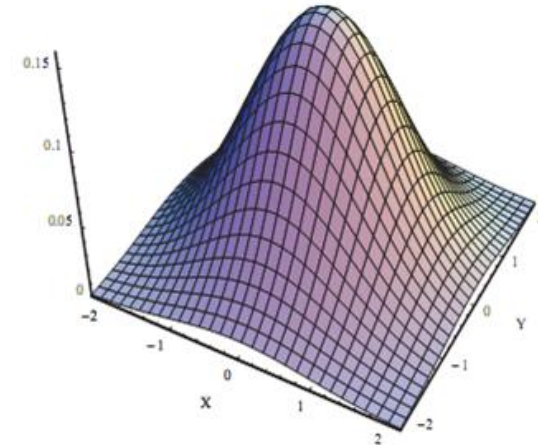
Weighted Smoothing



Pyramid



Cone



Gaussian

Task 2.3: Weighted smoothing

- The pyramid area function:

$$f(x, y) = -\alpha \cdot \max(|x|, |y|) + k$$

x and y designate the distance to the target pixel on the x and y axes, α is a parameter of the function and k a constant which you add for creating positive values.

Pyramid-area (5x5, $\alpha = 2$, $k=0$)

$$\begin{bmatrix} -4 & -4 & -4 & -4 & -4 \\ -4 & -2 & -2 & -2 & -4 \\ -4 & -2 & 0 & -2 & -4 \\ -4 & -2 & -2 & -2 & -4 \\ -4 & -4 & -4 & -4 & -4 \end{bmatrix}$$

Pyramide-are (5x5, $\alpha = 2$, $k=4$)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 2 & 4 & 2 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



Task 2.3

- The conical area function:

x and y designate the distance to the target pixel on the x and y axes, α is a parameter of the function and k a constant which you add for creating positive values.

$$f(x, y) = -\alpha \cdot \sqrt{x^2 + y^2} + k$$

Conic-area (5x5, $\alpha = 2$, $k=0$, exact)

$$\begin{bmatrix} -2\sqrt{8} & -2\sqrt{5} & -4 & -2\sqrt{5} & -2\sqrt{8} \\ -2\sqrt{5} & -2\sqrt{2} & -2 & -2\sqrt{2} & -2\sqrt{5} \\ -4 & -2 & 0 & -2 & -4 \\ -2\sqrt{5} & -2\sqrt{2} & -2 & -2\sqrt{2} & -2\sqrt{5} \\ -2\sqrt{8} & -2\sqrt{5} & -4 & -2\sqrt{5} & -2\sqrt{8} \end{bmatrix}$$

Conic-area (5x5, $\alpha = 2$, $k=2\sqrt{8}$, rounded)

$$\begin{bmatrix} 0 & 1 & 2 & 1 & 0 \\ 1 & 3 & 4 & 3 & 1 \\ 2 & 4 & 6 & 4 & 2 \\ 1 & 3 & 4 & 3 & 1 \\ 0 & 1 & 2 & 1 & 0 \end{bmatrix}$$



Task 2.4 : Laplacian Filters

- Laplacian filters are derivative filters used to find areas of rapid change (edges) in images.
- The Laplacian $L(x,y)$ of an image with pixel intensity values $I(x,y)$ is given by:

$$L(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

- Since images comprise of a set of discrete pixels, we calculate filters that approximate the second derivative

0	-1	0
-1	4	-1
0	-1	0

Details: <https://stackoverflow.com/questions/53544983/how-is-laplacian-filter-calculated>

